

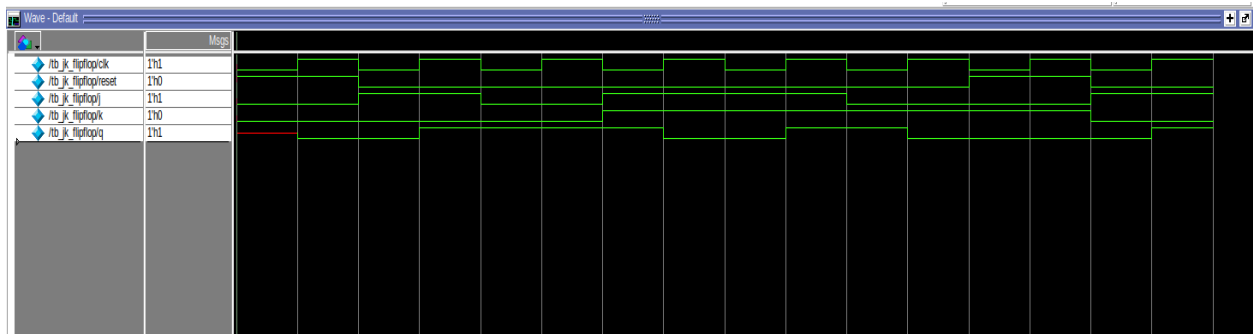
Design file

```
h /home/c4/jk_flipflop.v (/tb_jk_flipflop/ uut) - Default :
Ln#
1  module jk_flipflop(
2      input clk,
3      input reset,
4      input j,
5      input k,
6      output reg q
7  );
8
9      always @(posedge clk)
10     begin
11         if (reset)
12             q <= 1'b0;
13         else
14             case ({j, k})
15                 2'b00: q <= q;
16                 2'b01: q <= 1'b0;
17                 2'b10: q <= 1'b1;
18                 2'b11: q <= ~q;
19             endcase
20     end
21
22 endmodule
23
24
```

testbench

```
h /home/c4/tb_jk_flipflop.v (/tb_jk_flipflop) - Default :
Ln#
1  `timescale 1ns/1ps
2  module tb_jk_flipflop;
3      reg clk;
4      reg reset;
5      reg j;
6      reg k;
7      wire q;
8
9      jk_flipflop uut (
10         .clk(clk),
11         .reset(reset),
12         .j(j),
13         .k(k),
14         .q(q)
15     );
16     always #5 clk = ~clk;
17     initial begin
18         clk=0; reset=1;
19         j=0; k=0;
20         #10 reset=0;
21         j=1; k=0;
22         #10;
23         j=0; k=0;
24         #10;
25         j = 1; k = 1;
26         #10;
27         j = 1; k = 1;
28         #10;
29         j=0; k=1;
30         #10;
31         reset=1;
32         #10;
33         reset=0;
34         j=1;
35         k=0;
36         #10;
37         $finish;
38     end
39 endmodule
40
```

Waveform



```
cl4@CL4-45:~/f4pga-examples$ cd ~/f4pga-examples
cl4@CL4-45:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/.opt/f4pga
cl4@CL4-45:~/f4pga-examples$ export FPGA_FAM="xc7"
cl4@CL4-45:~/f4pga-examples$ source "$F4PGA_INSTALL_DIR/$FPGA_FAM/conda/etc/profile.d/conda.sh"
cl4@CL4-45:~/f4pga-examples$ conda activate xc7
(xc7) cl4@CL4-45:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/.opt/f4pga
(xc7) cl4@CL4-45:~/f4pga-examples$ export FPGA_FAM="xc7"
(xc7) cl4@CL4-45:~/f4pga-examples$ cd xc7
(xc7) cl4@CL4-45:~/f4pga-examples/xc7$
```

```
(xc7) cl4@CL4-45:~/f4pga-examples/xc7$ TARGET="arty_100" make -C jkff
```

```
(xc7) cl4@CL4-45:~/f4pga-examples/xc7$ lsusb
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 002: ID 046d:c077 Logitech, Inc. Mouse
Bus 001 Device 003: ID 046d:c34b Logitech, Inc. USB Keyboard
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
(xc7) cl4@CL4-45:~/f4pga-examples/xc7$ openFPGALoader --detect
unable to open ftdi device: -3 (device not found)
JTAG init failed with: unable to open ftdi device
```

```
(xc7) cl4@CL4-45:~/f4pga-examples/xc7$ find ~/f4pga-examples/xc7/jkff -name "*.bit"
```

```
(xc7) cl4@CL4-45:~/f4pga-examples/xc7$ openFPGALoader -b arty ~/f4pga-examples/xc7/jkff/build/arty_100/jk_flipflop.bit
unable to open ftdi device: -3 (device not found)
JTAG init failed with: unable to open ftdi device
(xc7) cl4@CL4-45:~/f4pga-examples/xc7$
```